

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Document Version:** 1.0
- Prepared by:** [Your Name], Senior Business Analyst
- Date:** [Current Date]
- Table of Contents**

1. **Introduction**

2. **Business Overview**

3. **Functional Requirements**

4. **Data Requirements**

5. **Technical Requirements**

6. **Assumptions and Dependencies**

7. **Testing and Validation**

8. **Acceptance Criteria**

1. Introduction

The purpose of this Business Requirements Document (BRD) is to outline the requirements for a data integration and aggregation project involving customer and order data. The project aims to load data from CSV files into delta tables, perform data cleansing, create summary tables, and aggregate data for customer spend analysis.

2. Business Overview

The business need for this project is to integrate customer and order data from CSV files into a unified data model, enabling analysis of customer spend behavior. The project will involve data loading, cleansing, transformation, and aggregation to create meaningful insights.

3. Functional Requirements

The following functional requirements have been identified:

1. **Data Loading**: Load customer and order data from CSV files into delta tables.
 - Source CSV files: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``
 - Target delta tables: ``customer`` and ``order``

2. ****Data Cleansing****: Remove null and duplicate records from the customer and order tables.
3. ****Data Transformation****: Join customer and order data using the `CustId` field and create a summary table (`ordersummary`).
 - Schema of `ordersummary` table: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`
4. ****SCD Type 2 Implementation****: Implement Slowly Changing Dimension (SCD) type 2 logic on the `ordersummary` table to track changes to customer data.
 - Update `StartDate` and `EndDate` columns accordingly
5. ****Data Aggregation****: Aggregate data from the `ordersummary` table to create a new table (`customeraggregatespend`) with columns `Name`, `TotalAmount`, and `Date`.
6. ****Data Loading into `customeraggregatespend`****: Load aggregated data into the `customeraggregatespend` table.

4. Data Requirements

The following data requirements have been identified:

- **Customer Data****:
 - + Source: CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`
 - + Schema: `CustId`, `Name`, `EmailId`, `Region`
- **Order Data****:
 - + Source: CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`
 - + Schema: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`
- **Order Summary Data****:
 - + Target table: `ordersummary`
 - + Schema: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`
- **Customer Aggregate Spend Data****:
 - + Target table: `customeraggregatespend`
 - + Schema: `Name`, `TotalAmount`, `Date`

5. Technical Requirements

The following technical requirements have been identified:

- **Data Storage****: Use Databricks delta tables for storing data.

- Data Processing**: Use Databricks for data processing and transformation.
- Catalog and Schema**: Use `catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdic\_wizard"` for storing tables.

6. Assumptions and Dependencies

The following assumptions and dependencies have been identified:

- Data Quality**: The source CSV files are assumed to be in the correct format and contain valid data.
- Databricks Environment**: The Databricks environment is assumed to be set up and configured correctly.

7. Testing and Validation

The following testing and validation activities will be performed:

- Data Validation**: Validate data loaded into delta tables against source CSV files.
- Data Transformation**: Validate data transformation logic and output.
- SCD Type 2 Implementation**: Validate SCD type 2 logic and output.

8. Acceptance Criteria

The following acceptance criteria have been defined:

- Data Loading**: Data is loaded correctly into delta tables.
- Data Cleansing**: Null and duplicate records are removed from customer and order tables.
- Data Transformation**: Data is transformed correctly into the `ordersummary` table.
- SCD Type 2 Implementation**: SCD type 2 logic is implemented correctly on the `ordersummary` table.
- Data Aggregation**: Data is aggregated correctly into the `customeraggregatespend` table.

By meeting these acceptance criteria, the project will be considered successful.